Type MS · Identify urinary system and nephron structures

Answer 1 of 9 · Question 6 part a · 4 marks

2023 NOV · Paper 01 · extract 1 of 2 · find part a in this mark scheme extract

Number				
6	(a)	L = Bowman's capsule; M = glomerulus; N = collecting duct; 0 = loop of Henlé ;		4
	(b) (i)	• ultrafiltration; • high blood pressure in B/glomerulus; • forces fluid/water into L/Bowman's capsule/out;		3
	(ii)	• blood cells/named; • proteins;		2
	(iii)	line labelled G pointing to proximal convoluted tubule;		1
	(c)	any five from • more sweat; • hypothalamus detects decreased water potential of blood; • ADH secreted; • increases permeability of collecting duct; • causes more reabsorption of water; • volume of filtrate reduced; • concentration of filtrate/urine increased;		5

Total 15 marks

Type MS · Identify urinary system and nephron structures

Answer 1 of 9 · Question 6 part a · 4 marks

2023 NOV · Paper 01 · extract 2 of 2 · find part a in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Identify urinary system and nephron structures

Answer 2 of 9 · Question 8 part a.i · 2 marks

2020 JAN · Paper 01 · find part a.i in this mark scheme extract

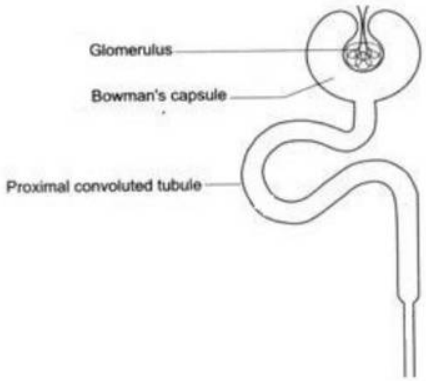
Number				
8	(a) (i)	X = urethra (1) Y = ureter (1)		2
	(ii)	Any two from: • transports oxygenated blood from the heart (1) • to the body/named organ (1) • maintains high blood pressure (1)		2
	(iii)	Any two from • blood in A/renal artery contains more oxygen/oxygenated (1) • blood in A/renal artery contains more urea (1) • blood in A/renal artery contains more ions (1) • similar concentration of glucose/B renal vein has less glucose (1) • blood in A/renal artery contains less carbon dioxide (1)	Allow reverse argument for B/renal vein	2
	(b) (i)	Any four from: • dialysis fluid/dialysate contains a low(er) concentration/no urea/waste products/ions (1) • urea/waste/ions diffuse from blood (1) • through dialysis membrane (1) • from high(er) concentration (in the blood) to a low(er) concentration (in the dialysate)/down a concentration gradient (1) • dialysis membrane is partially-permeable(1)	ORA allow selectively permeable membrane	4
	(ii)	• to prevent blood from clotting whilst in the machine (1) • (added as it enters the machine) to prevent clots from blocking the machine (as blood flows through) (1) • (not added at the end) as blood clotting properties need to return to normal before passing back into patient (1)		3

Total for question 8 = 13 marks

Type MS · Identify urinary system and nephron structures

Answer 3 of 9 · Question 9 part b · 4 marks

2020 NOV · Paper 01 (regional R) · extract 1 of 2 · find part b in this mark scheme extract

9 (a) (i)	• evidence of 23500 – 4000 (= 19500) (1) • 19500 ÷ 4000 (x 100)(1) • 488(%)	Allow full marks for correct final answer. Allow 487.5(%) Allow ecf if incorrect answer from mp1 – max 2 marks	3
(ii)	other lifestyle factors/genetic predisposition could have an influence/named lifestyle factor(1)	Allow gender/age	1
(b)	A diagram similar to:  	One mark for diagram. One mark for each correct label.	4
(c)	any four from • ultrafiltration does not take place(1) • in glomerulus(1) • less/no waste products/named waste product/water/glucose pass through to tubule(1) • greater volume of waste in blood/more water in blood(1) • glucose/ions not reabsorbed(1) • glucose found in urine(1) OR any four from • blood not filtered effectively/ultrafiltration ineffective(1) • in glomerulus (1)		4

Type MS · Identify urinary system and nephron structures

Answer 3 of 9 · Question 9 part b · 4 marks

2020 NOV · Paper 01 (regional R) · extract 2 of 2 · find part b in this mark scheme extract

	• larger molecules/named larger molecules pass through to tubule/Bowmans capsule(1) • not all glucose/ions reabsorbed(1) • glucose/protein found in urine/urine more concentrated(1)		

Total for Question 9 = 12 marks

Type MS · Identify urinary system and nephron structures

Answer 4 of 9 · Question 1 part a.i · 3 marks

2022 JAN · Paper 02 · extract 1 of 2 · find part a.i in this mark scheme extract

1	(a)	(i)	• first convoluted tubule; • loop of Henlé; • second convoluted tubule;	diagram only, no labels required 3
		(ii)	44mm; 44 ÷ 650; = 0.068mm;	ecf from mp1 max 2 marks Allow 0.0676 3
		(iii)	• protein; • red/white blood cells/platelets;	2
		(iv)	glucose;	1
	(b)	(i)	Any 2 from: • same person; • same activity levels; • same volume of water; • same food eaten;	2
		(ii)	710 ÷ 4; = 178 cm^3 per hour;	2 Allow 1 mark if division by 4 is shown Allow 177.5
		(iii)	• less urine production/lower urine loss on day two; • more ADH secreted on second day; • more water reabsorbed; • more water lost by sweating;	Allow reverse argument for all marking points 3

Total 16 marks

Type MS · Identify urinary system and nephron structures

Answer 4 of 9 · Question 1 part a.i · 3 marks

2022 JAN · Paper 02 · extract 2 of 2 · find part a.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Identify urinary system and nephron structures

Answer 5 of 9 · Question 7 part a.i · 1 mark

2022 JUN · Paper 02 (regional R) · extract 1 of 2 · find part a.i in this mark scheme extract

7 (a) (i)	C; (ureter) *A is a blood vessel* *B is a blood vessel* *D is not attached to kidney*		1
(ii)	A; (bladder) *B is part of the reproductive system* *C is part of the reproductive system* *D leads from the bladder*		1
(b)	• more oxygen; • less carbon dioxide; • more urea;	accept reverse for blood leaving kidney	3
(c) (i)	16.8 - 3.6 = 13.2; 13.2 ÷ 3.6 =3.66; 3.66 × 100 = 366.7%;		3
(ii)	• urea; • large/significant increase in protein-rich diet;		2
(iii)	Any four from • urea, uric acid and ammonia masses are different between the two diets; • ref. creatinine; • amount same/nearly same; • in both diets; • possible result of muscle breakdown;		4

Total 14 marks

Type MS · Identify urinary system and nephron structures

Answer 5 of 9 · Question 7 part a.i · 1 mark

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Type MS · Identify urinary system and nephron structures

Answer 6 of 9 · Question 7 part a.ii · 1 mark

2022 JUN · Paper 02 (regional R) · extract 1 of 2 · find part a.ii in this mark scheme extract

7 (a) (i)	C; (ureter) *A is a blood vessel* *B is a blood vessel* *D is not attached to kidney*		1
(ii)	A; (bladder) *B is part of the reproductive system* *C is part of the reproductive system* *D leads from the bladder*		1
(b)	• more oxygen; • less carbon dioxide; • more urea;	accept reverse for blood leaving kidney	3
(c) (i)	16.8 - 3.6 = 13.2; 13.2 ÷ 3.6 =3.66; 3.66 × 100 = 366.7%;		3
(ii)	• urea; • large/significant increase in protein-rich diet;		2
(iii)	Any four from • urea, uric acid and ammonia masses are different between the two diets; • ref. creatinine; • amount same/nearly same; • in both diets; • possible result of muscle breakdown;		4

Total 14 marks

Type MS · Identify urinary system and nephron structures

Answer 6 of 9 · Question 7 part a.ii · 1 mark

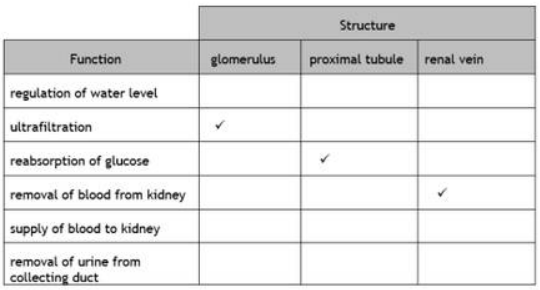
2022 JUN · Paper 02 (regional R) · extract 2 of 2 · find part a.ii in this mark scheme extract

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Type MS · Identify urinary system and nephron structures

Answer 7 of 9 · Question 7 part a.ii · 3 marks

2023 JAN · Paper 02 · extract 1 of 2 · find part a.ii in this mark scheme extract

Number					
7	(a)	(i)	Hypothalamus/osmoregulatory centre;		1
		(ii)		Reject more than one tick in any column	3
	(b)	(i)	• breakdown of proteins/amino acids/deamination/removal of NH_2/amine group; • in the liver;		2
		(ii)	Five from: • second sample has less water/more urea/other components; • urine is more concentrated/less urine produced; • person is less hydrated/less water/lower water potential in the blood; • ADH has been produced; • increased the permeability of the collecting duct; • more water has been reabsorbed (by the blood);		5

Total 11 marks

Type MS · Identify urinary system and nephron structures

Answer 7 of 9 · Question 7 part a.ii · 3 marks

2023 JAN · Paper 02 · extract 2 of 2 · find part a.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Identify urinary system and nephron structures

Answer 8 of 9 · Question 6 part b.iii · 1 mark

2023 NOV · Paper 01 · extract 1 of 2 · find part b.iii in this mark scheme extract

Number				
6	(a)	L = Bowman's capsule; M = glomerulus; N = collecting duct; 0 = loop of Henlé ;		4
	(b) (i)	• ultrafiltration; • high blood pressure in B/glomerulus; • forces fluid/water into L/Bowman's capsule/out;		3
	(ii)	• blood cells/named; • proteins;		2
	(iii)	line labelled G pointing to proximal convoluted tubule;		1
	(c)	any five from • more sweat; • hypothalamus detects decreased water potential of blood; • ADH secreted; • increases permeability of collecting duct; • causes more reabsorption of water; • volume of filtrate reduced; • concentration of filtrate/urine increased;		5

Total 15 marks

Type MS · Identify urinary system and nephron structures

Answer 8 of 9 · Question 6 part b.iii · 1 mark

2023 NOV · Paper 01 · extract 2 of 2 · find part b.iii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Identify urinary system and nephron structures

Answer 9 of 9 · Question 5 part a.ii · 1 mark

2025 JUN · Paper 01 (regional R) · extract 1 of 2 · find part a.ii in this mark scheme extract

5	(a)	(i)	• flows from artery; • where blood is at high pressure; • into vein; • because blood is at a low pressure;	mp1- accept in context of artery leaving heart mp3- accept in context vein of returning to the heart If comparative answer given for pressure score marking point 2 and 4	4
		(ii)	glomerulus;		1
		(iii)	• large (surface) area; • for diffusion / molecules/ named molecules to pass out(of blood);	More area for diffusion = 2 marks	2
		(iv)	more urea;		1
	(b)		• no/less gradient between liquid X and blood (glucose)/blood and X have same concentration/eq; • glucose does not pass out/ glucose is not lost: • by diffusion • needed for respiration; • minimises osmotic effect;		3
	(c)		• protein digested/broken down into <u>amino acids</u>; • (excess) **<u>amino acids</u>** broken down/deaminated; • urea; • less protein means less urea produced; • less urea means less to be filtered/removed ; • high concentration of urea/nitrogenous waste can be toxic;	MP3– accept urea from protein breakdown	4
					Total =15

Type MS · Identify urinary system and nephron structures

Answer 9 of 9 · Question 5 part a.ii · 1 mark

2025 JUN · Paper 01 (regional R) · extract 2 of 2 · find part a.ii in this mark scheme extract

Question number	Answer	Notes	Marks